Pay and Performance on the Ice: Revisiting the Principal-Agent Problem in Hockey

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Abstract
The objective of this paper is to explore the effects of time on ice, salary, and contract properties on the performance of NHL athletes. Optimal contract design is central to both management and players, as teams seek to minimize risks, while players aim to maximize long-term security. Speculation on why player output might decline in academia, industry, and popular discourse remains active since performance measurements fluctuate. Using Ordinary Least Squares regression with fixed-effects specifications on 1089 NHL forwards in 3742 observations across 11 seasons, I test if the agent (player) performance output on behalf of the principal (manager) declines or improves. I find no evidence of shirking or incentive behavior from contract boundary years. Salary and time-on-ice remain strong indicators of performance output, supporting claims that modern contracts do not induce changes to performance.
Keywords: NHL, xgar, principal-agent, contracts

Introduction

In the National Hockey League, a declining Jonathan Huberdeau is believed to be an "overpaid" athlete (or having a bad contract), with a salary of $10.5 million over 8 years and output below expectations. Once that contract was signed and he moved to a new team, his output declined. Whether Huberdeau's decline reflects a moral hazard, diminishing returns to ice time, or measurement error in contract valuation remains empirically unresolved. Major sports leagues like the NHL (together with Major League Baseball, the National Football League and the National Basketball Association) generate tens of billions of dollars in tax revenue, causing immense economic stimulus locally and nationally. Management of these teams balances efficient contracts with building a winning composition, while players seek security and the prestige that a large contract produces. The economic impact increases as firms from other sectors employ the top level athletes for marketing purposes, and there are significant economic effects in the cross-over between sports, markets, and politics. While profit maximization is vital for the manager and the player occupying the negotiating table, it is winning championships that matters the most.

The circumstances of negotiations between player and manager can be viewed as a Principal-Agent problem(Purcell 2009, Wu 2020). I define this using as: "...a relationship in which one party, known as the principal, contracts with another party, known as the agent, in the expectation that the agent will act on behalf of the principal."(McEachern 2009). In hockey, the principals are general managers representing the organization and the agents are the players (or representatives of) acting on behalf of the organization, on the ice. The principal and agent are both in a highly competitive market and engage in competitive games. Examples of a moral hazard might be after securing high salary and security, a forward shirks his xGAR/60 output by changing off-season training or in some way sees lower output. However, adverse selection would be a player with a history of injury or issues seeking same-value contracts as forwards without the risks. An information asymmetry could exist if a player is beginning to feel the effects of aging[1] on his endurance, recovery, or other parts of output and his output falls, but management is not privy to any information. Information asymmetry is often a precursor that leads to the first two, and this could lead to both a moral hazard or an adverse selection. I look to test if these manifest in athletes shirking after achieving new contract (year one), react to incentives to overperform (final year), how the point of time in the contract interacts with on-ice playtime, or if higher salary yields higher output.

This paper will be structured as follows: first, I review some recent literature on pay and

[1]This would be a likely unobservable characteristic.

performance metrics, hockey contracts, and hockey performance statistics. It will show that work thus far can be approached from a unique vector like the models I propose. Second, I will explain my data, the hypotheses to be tested, and my regression model for each hypothesis. The third will summarize the results, and final section will be my conclusion revisiting the hypotheses, underlying theory, and implications of this analysis.

Literature Review

The contemporary literature on pay determination, incentives, contract structure, and athlete performance has greatly advanced on the analytical side of the NHL. Statistical analytics continues to evolve as hobbyists and professionals commit themselves to evolving its application to sports. I focus on research related specifically to hockey, but my review of the literature begins with glass workers.

Lazear (2000) did not conduct a sports study, but used a data set of glass workers at a sample size of 3000 to look at the effect of financial incentives on worker behaviour. This creates an important baseline for pay and performance in economic analysis. In this study, his method was to look at the data before and after the change in pay. In one word, Lazear's work demonstrates that piece-rate compensation mitigates moral hazard by aligning worker incentives with output. The effect was significant because it increased productivity by 20-36% (Lazear, 2000, p.1). The inverse of this relates to the study: workers who have security in their wages may see a drop in output in the glass factory. (Lazear, 2000 p.4) Lazear suggests workers may withhold effort under fixed wages if they fear that exceeding expectations will permanently raise performance benchmarks in a 'ratchet effect'. This pay and performance work helps conceptualize how a hockey player might change their output, but doesn't answer what model is best to use.

Rob Simmons (2022) explores the problem of pay and performance in his work "Professional Labor Markets in the Journal of Sports "Economics". Simmons identifies important features of sports labor markets that relates to the NHL: teams exercise monopsony power due to limited buyer competition[2], rent from scarce ability is shared between player and team, and (less importantly) bargaining on pay outcomes does result in a solution via Solow and Krautman's Nash bargaining model (Simmons, 2022, p.729). This identifies the time-invariant properties of players and teams. Simmons finds an Ordinary Least Squares model using variables of interest with fixed effects to be a dominant estimation method for sports. Salary amounts are introduced as log values of their original values (to normalize) while there is typically a concave curve for player marginal productivity because of 'wear and tear'

[2] Only 32 NHL teams

(Simmons, 2022, p.731), and performance measurements are not merely something as simple as Points or points per game.

Purcell (2009), in "Long-Term Contracts and the Principal-Agent Problem", measures performance using total points, a metric that fails to capture defensive contributions, ice time quality, or situational context. Points alone cannot capture a player's comprehensive contribution to winning, as they omit defensive play, penalty discipline, and context-dependent output. Purcell found that there was evidence of shirking in sports, "...security may lead a player to lower his effort level" (Purcell, 2009, p.48). However, her results 'indicate a statistically significant negative relationship between signing a long-term contract and performance in the subsequent year' (Purcell, 2009, p.63). Using such a narrow measurement like points output alone fails to capture the picture of how the player contributed that year.

In Katarina Wu's 2020 honours thesis, she tested a point-per-game model that explored contract and performance. Gannon (2009) uses a similar model in testing contract boundary years. In her work, Wu used a point per game metric as the dependent variable when she regressed it on variables of interest with fixed effects. This method aims to address concerns with Purcell's use of simple point totals. This paper follows Wu and Gannon, but employs my own hypotheses with models using panel data employing fixed effects, and an outcome variable that is itself an aggregate composite variable that captures defensive play, penalty discipline, and any context-dependent output.

Data

No single public database provides skater biographical, contract, and performance data, which required manual merging from multiple sources. Creating the data set meant using several sources and combining them using Excel VBA scripts and Stata methods. The contract data for this were sourced from the Spotrac database, using their web forms specified to seasons 2014-2015 to 2024-2025. I constructed new variables for contract boundary and contract state using the variables[3] from Spotrac's database. This was chosen because the distribution of contracts skewed heavily towards deals that were one to three years in length. The after-tax average annual value, or AAV, featured leptokurtic distribution with a right positive skew. I apply a log transformation to reduce right skewness and approximate log-normality, consistent with standard practice in wage regressions (Wu 2020; Mincer 1974). The contract values were multiplied by the applicable effective tax rate for that year, and then a natural log was taken. The original observations had forwards, defence and goalies; for this paper, I narrowed it down to forwards (center, left wing and right wing). These

[3] Contract start, contract end, length

position variables will be controlled for as time-invariant effects by the fixed effects terms. The distribution of contracts can be seen in Figure 1, and it is skewed towards shorter contracts in part because management excels at identifying high performing players who earn longer contracts, and rental (short term) players who fulfill depth roles. Selection bias is, while unlikely, possible because data cleaning means that seasons and contracts outside of this period are not included. [4]

Performance statistics for the outcome variable came from the Evolving Hockey's Expected Goals Above Replacement skater tables, which I will define and explain in the model section. Additional information on skater measurements includes games played, time on the ice for all situations, and age. I selected the seasons 2014-2015 to 2024-2025 for the time window, providing 11 seasons of data. After merger with the contract data there were 4844 total observations, and this was trimmed down to 3742 by focusing on observations that had both performance and contract data. This represents a total of 1089 unique players over different periods of time.

Regarding the summary statistics of this outcome variable and the variables of interest that I focus on in this paper, there are some important data-points to highlight. The minimum and maximum of xGAR per 60 $(-2.419, 2.072)$ highlights that the value can go negative, as a result of it being a composite. The mean approaching zero may indicate that there is some sort of parity of goals being scored for and against, but the standard deviation is large considering the minimum and maximum values. Whether or not this is meaningful or results from someone having to lose is unclear. Age shows that the National Hockey League requires experience with a mean age of 26, where the minimum is 18 (draft year). Games played reveal that the average is not playing a full 82 game season, rather approximately 64% of a full season. These data created the source on which the regression models were run.

Regression Model

The purpose of these regression models is to see whether salary, contract timing, and time on ice help explain variation in player performance once obvious confounders are controlled for. I perform estimations with 3 related OLS regression models. Here, i indexes players and s seasons:

Null Hypothesis 1 - Salary and time on ice have no effect on performance output.

Hypothesis 1 - salary and time on ice are associated with performance, even after basic experience controls:

[4] Three observations with implausible salary values (likely data entry errors) were excluded.

Model 1:

$$\begin{aligned}\text{xGAR}_{60,is} = \beta_0 &+ \beta_1 \ln(\text{AAV}_{is}) + \beta_2(\text{TOI}_{60,is}) \\ &+ \beta_3\text{Age}_{is} + \beta_4\text{Age}^2_{is} + u_{it}\end{aligned}$$

xGAR_{60} is expected goals above replacement per 60 minutes; ln(AAV) is the natural log of the average annual salary after tax; (TOI) is the total minutes played per 60 minutes; age and age squared.[5] I expect $\beta_1 > 0$ if higher-paid players are, on average, better, and $\beta_2 > 0$ if better players earn more ice time.

Null Hypothesis 2 - Log-salary, time on ice, and the interaction between contract year and time on ice have no effect on performance.

Hypothesis 2 - One Plus FE: This expands on hypothesis one in an alternate form. It includes values representing how many years into a contract (and years left) with fixed effects. The player fixed effects control for time-invariant ability, and the team fixed effect will absorb coaching quality, roster, and venue effects.

Model 2:

$$\begin{aligned}\text{xGAR}_{60,it} = \beta_1 \ln(\text{AAV}_{it}) &+ \beta_2(\text{TOI}_{60,it}) + \beta_3\text{Age}_{it} \\ &+ \beta_4\text{Age}^2_{it} + \beta_5(\text{YearsInto}_{it} \times (\text{TOI}_{60,it})) \\ &+ \beta_6(\text{YearsLeft}_{it} \times (\text{TOI}_{60,it})) + \alpha_{it} + \gamma_{it} + u_{it}\end{aligned}$$

YearsInto counts the seasons since the contract started and YearsLeft counts the remaining seasons. α_{it} are player fixed effects and γ_{it} are team fixed effects. This design compares each player with himself across contract years, while keeping team context constant. Under a shirking story, the time on ice would be less productive early in contracts (negative β_3); under an incentive story, the last years would look different from the rest. Standard errors are clustered by player.

Null Hypothesis 3 - Log salary, icetime, and contract boundary years have no effect on performance.

Hypothesis 3 - the first and last years of a contract differ from the middle years. The third specification is the following.

Model 3:

$$\begin{aligned}\text{xGAR}_{60,it} = \beta_1 \ln(\text{AAV}_{it}) &+ \beta_2(\text{TOI}_{60,it}) + \beta_3\text{Age}_{it} \\ &+ \beta_4\text{Age}^2_{it} + \beta_5(\text{FirstYear}_{it}) + \beta_6(\text{FinalYear}_{it}) \\ &+ \alpha_{it} + \gamma_{it} + u_{it}\end{aligned}$$

FirstYear equals 1 in the first season of a contract. FinalYear equals 1 in the last year of a

[5] Age is squared because of the concavity of the value over time; as a player gets older, output rises but plateaus and then declines.

contract. I took the natural log of salary (AAV) to approximate its normal distribution. The time on ice is adjusted to the average per 60 minutes (per game). I again use player season fixed effects α_{it} but switch to season fixed effects in γ_{it} here to test time-invariant things like changes to Salary Ceiling, rule changes, etc. A joint F-test on β_2 and β_3 asks whether being in a start or end year matters at all. As before, the standard errors are clustered by player.

Outcome Variable

For these regression models, I chose Evolving Hockey's Expected Goals Above Replacement (xGAR) as my outcome variable, to aggregate important performance data into a single value. The explanation provided by Evolving Hockey further supports the usage of this as an outcome variable: it can comprehensively represent a player's contributions to their team, it uses goals as a foundation for the method, it treats rookies and veterans the same, and the method allows for analysis of the internal components individually. It is all grounded in Baseball Sabermetrics (simply a term representing empirically-driven statistical analysis of sports output) Wins Above Replacement. Although xGAR does not fully capture teammate effects (goal winning, defense roles), it represents a substantial improvement over the point-based metrics used in previous studies.

$$xGAR = xGAREV_o + xGAREV_d + xGARPP_O + xGARSH_D + xGARP_T + xGARP_D$$

In the xGAR composite, there are composite variables that represent goals above the replacement Offensive, Defensive, and Penalty performance statistics. It compares these - in the Above Replacement forms - against a theoretical replacement player. Each of are themselves a composite value that represents a player's individual performance at even strength, power play, shorthanded, with penalty taken and drawn in mind. It is important to explain because, despite the moving parts inside of xGAR, the entire idea of Goals Above Replacement is predicated on the idea of expected goals. The expected goals are a synthesis of the distance, angle, and type of unblocked shots. Expected Goals Above Replacement takes this simplistic measurement of data, and using weighted values for its inputs, creates a singular number: for every increase in the variables of interest, a higher or lower xGAR measures a player's on-ice contributions to their team winning.

Variable of Interest

The primary variables of interest are the Average Annual Value of a hockey player's contract, their per 60 minutes on the average of ice playtime, and the properties of their contract. This AAV is the amount agreed on between the Principal (GM) and the Agent (player), and serves as the main variable of interest. There is a potential simultaneous causality between the time on ice measurement and the outcome variable. The better player you are (outcome variable), the more ice time you get (variable of interest). The more ice time you get, the higher the outcome xGAR can reach.

Results

In the baseline OLS (no fixed effects) for the first hypothesis, log salary is not statistically significant ($\beta = -0.024$, $p = 0.059$) but borders on it, while time-on-ice per 60 has a strong positive effect ($\beta = 3.601$, $p < 0.01$). Age-squared is negative and significant ($\beta = -0.00085$, $p = 0.001$), consistent with a concave aging curve. As a player ages, accumulating experience, there is a plateau followed by an eventual decline. The model explains about 18% of the variation in performance, and the model appears too naive without fixed effects.

Once fixed effects for players and team or season are introduced, the log salary coefficient remains negative but becomes highly significant ($\beta = -0.0922$, $p < 0.001$). The negative salary coefficient in the Fixed Effects models reflects that within-player salary-increases occur later in a player's career when age-induced decline is significant, rather than salary causing lower performance.. Although the time on ice per 60 itself is significant, it might be simultaneously causal with xGAR: the greater the ice time, the greater the chance for performance output, and the greater the performance per 60, the greater the ice time they earn from the coach. The interaction of years into and years left of a contract with time on ice per 60 minutes (β_5 = -0.0262, p¡0.459; β_6 = 0.0009, p ¡0.969) is not statistically significant. Age (β_3 = 0.0414, $p < 0.006$) is borderline significant, and age (β_4 = -0.001, $p < 0.01$) squared is understandably significant. This is only one way to look at contracts and does not isolate the first or final years like the final model.

In the contract boundary-year model, the logarithmic salary remains negative and significant (β_1 = -0.097, $p < 0.001$). The first-year and final-year dummies are small and statistically insignificant (0.021 and 0.005, respectively), and a joint test fails to reject the null that both equal zero ($F = 0.97$, $p = 0.38$). This does not provide evidence of systematic shirking or contract-year effort spikes. These results do speak to the status of OLS assumptions.

The OLS assumptions do not hold for the first model, but hold for the second two models. The first model is a simple or naive model to test Hypothesis 1, and the omission of variables accounting for ability[6] not only leads to a bias, but also results in a positive correlation between logarithmic salary and the error term. While the second assumption fails in the first, the introduction of clustering for the Fixed Effect player_id handles this exception. The third assumption holds in all three because there is no pattern of outliers. An omitted variable bias is observed in these models: a positive bias in the first and a negative bias in the other two. Since the first has no account for team wide or player time-invariant properties. Without this, a high logarithmic salary and ice time incorrectly explain the higher xGAR output. The third model attempted to control for rule changes, venue changes, or other seasonal time-invariant shocks. In the last two, variables such as injury change over time and are not included. A final potential omitted variable could be the quality of line-mates and the quality of opponents. The results of these models tell a story that may not wield predictive power in their current form, but nonetheless account for increasing variation.

[6] The ability of players, coaches, and teams.

Conclusion

In each of the three models, a different approach was used to look at what effect would a unit increase of logarithmic after-tax salary, time-on-ice, or contract properties have on a player's performance output. This engages with the Principal-Agent problem by looking at those three variables of interest and their effect on performance. It tests whether a ratchet effect could be observed in the salary of players, if adverse selection can be observed through ice time, and if shirking or incentives can be seen in via different contract properties.

The three models tell a consistent story regarding shirking: there is no evidence that players reduce effort in the middle of contracts or suddenly increase effort when contracts are close to ending. The first-year and final-year dummy variables are small and jointly insignificant ($F = 0.97$, $p = 0.38$). The negative and significant coefficient on log salary in the fixed effects models is counterintuitive but likely reflects that within-player salary increases coincide with aging and natural performance decline rather than any causal effect of pay on effort. The Time-on-ice per 60 is the most robust predictor of xGAR/60 in all specifications ($\beta \approx 3.0$–3.7, $p < 0.01$), confirming that players who receive more playing time produce more value. The negative coefficient on age squared supports a concave experience curve: performance rises to a plateau, then declines. The negative coefficient between players on salary likely reflects that pay increases the lag in peak performance rather than any causal effect of pay on effort.Although R^2 itself is not low at¿ 50% for the fixed effects model, the relatively low values of within-R^2 (around 3-6%) indicate substantial unexplained variation.

If Simmons is correct that fixed-effects OLS models, such as these high dimension ones are best, it would mean that there are further omitted variables, unobserved data, or more time-invariant effects to be included in these models. Composite outcome variables are still likely to be the best option in order to capture a full picture of what transpires on the ice. Despite shortcomings, weak predictive power and low variance explanation the models do not find shirking, incentive shocks, or other problems. Information asymmetries may be minimal or outliers. A principal relying on an agent to act on the behalf on the hockey rink appears well-founded as the agent appears aligned with the principal[7]. Within the limits of this sample and design, modern NHL contracts do not generate systematic shirking or incentive effects; instead, performance differences are driven by the underlying ability and how much a player is actually used on the ice. Future research should incorporate injury data, locker room variables, off ice training, and the potential for piece-rate incentives.

[7] A principal and agent are aligned in their vision for winning and profits

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Appendix

Table 1: Performance (xGAR/60)

	(1) Hypothesis 1	(2) Hypothesis 2	(3) Hypothesis 3
Log Salary	-0.024*	-0.092***	-0.096***
	(0.013)	(0.023)	(0.021)
Time-on-Ice per 60	3.601***	3.063***	2.874***
	(0.199)	(0.321)	(0.311)
Age	0.041***	0.093***	0.000
	(0.015)	(0.029)	(.)
Age Squared	-0.001***	-0.002***	-0.002***
	(0.000)	(0.001)	(0.001)
Years Into Contract × Time-on-Ice per 60		-0.026	
		(0.035)	
Years Remaining × Time-on-Ice per 60		0.001	
		(0.023)	
1st year			0.021
			(0.015)
Final year			0.005
			(0.020)
Constant	-0.844***	-0.324	2.377***
	(0.229)	(0.384)	(0.596)
R^2	0.180	0.504	0.493
Adj. R^2	0.179	0.346	0.337
Within R^2		0.077	0.062
F.E.	No	Yes	Yes

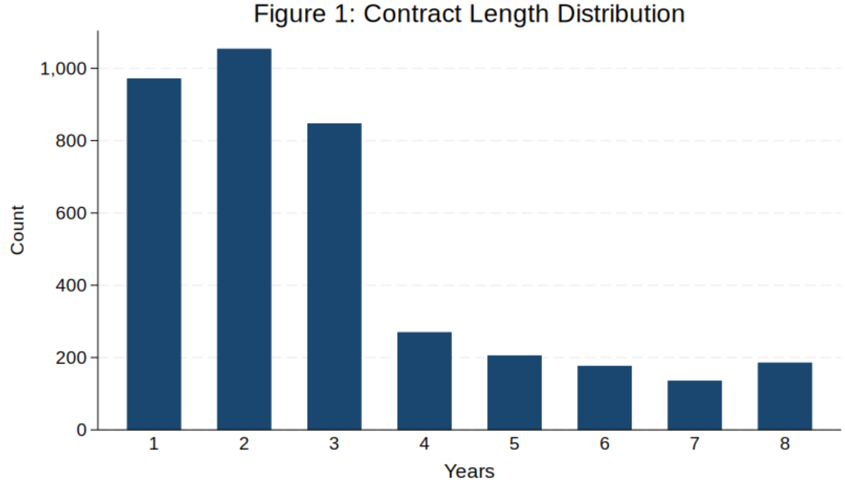
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

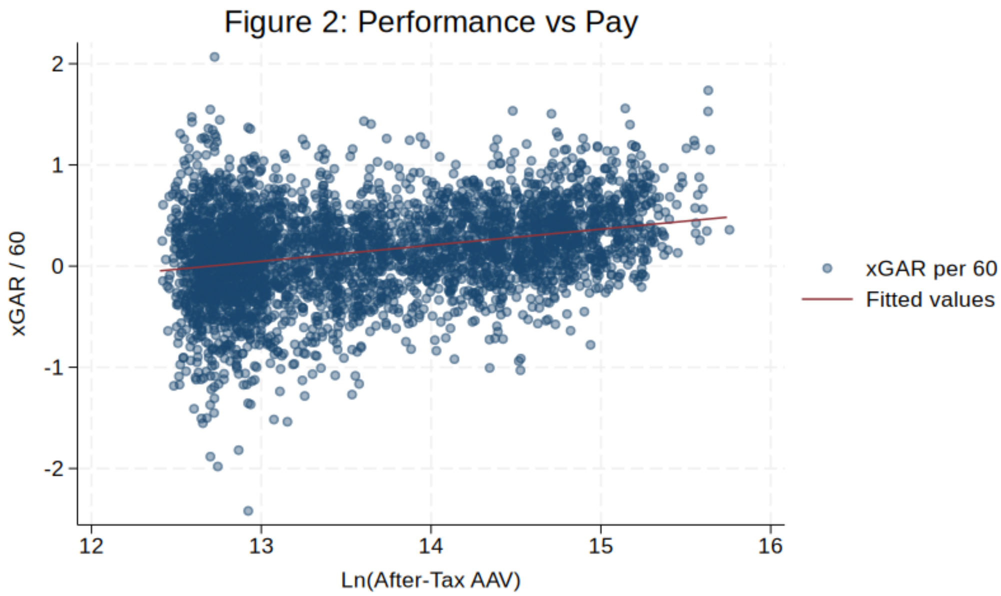
Table 2: Descriptive Statistics

	Mean	SD	Min	Max
xGAR per 60	0.150	0.453	-2.419	2.072
Log Salary	13.635	0.851	12.403	15.742
Time-on-Ice per 60	0.231	0.057	0.085	0.381
Games Played	52.317	25.487	3.000	82.000
Age	26.567	4.281	18.000	45.000
Age Squared	724.140	239.216	324.000	2025.000
Years Into Contract	1.594	0.828	1.000	4.000
Years Remaining	1.437	1.707	0.000	8.000
Final year	0.373	0.484	0.000	1.000
First year	0.591	0.492	0.000	1.000

* Outcome Variable and Variables of Interest

Figure 1: Contract Length Distribution





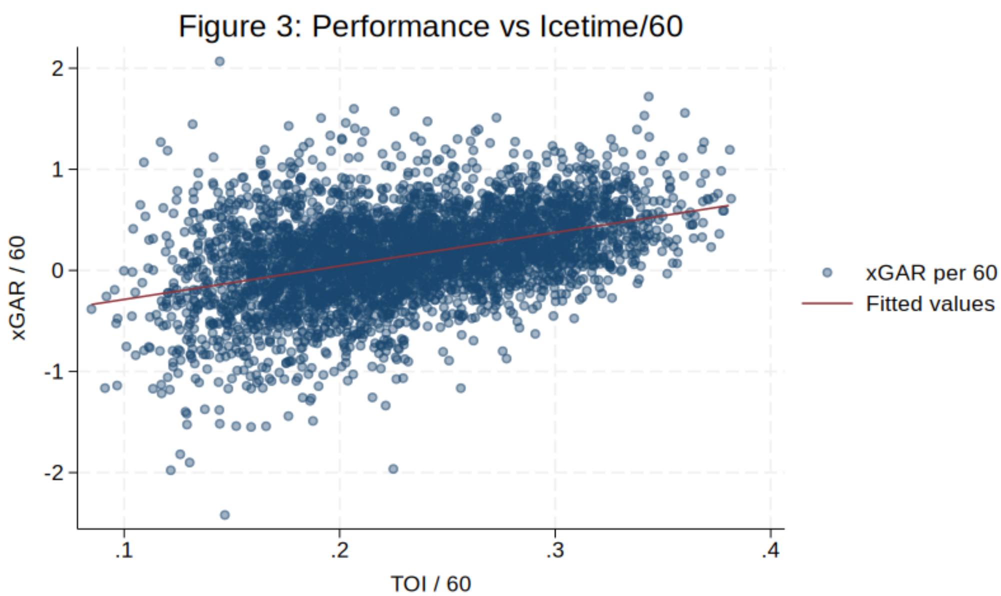


Figure 1: Performance vs Ice-time/60